

- catalyst** (kat'-uh-list) A chemical agent that selectively increases the rate of a reaction without being consumed by the reaction.
- cation** (cat'-i'-on) A positively charged ion.
- cation exchange** (cat'-i'-on) A process in which positively charged minerals are made available to a plant when hydrogen ions in the soil displace mineral ions from the clay particles.
- cecum** (sē'-kum) (plural, **ceca**) The blind pouch forming one branch of the large intestine.
- cell** Life's fundamental unit of structure and function; the smallest unit of organization that can perform all activities required for life.
- cell body** The part of a neuron that houses the nucleus and most other organelles.
- cell cycle** An ordered sequence of events in the life of a cell, from its origin in the division of a parent cell until its own division into two. The eukaryotic cell cycle is composed of interphase (including G₁, S, and G₂ phases) and M phase (including mitosis and cytokinesis).
- cell cycle control system** A cyclically operating set of molecules in the eukaryotic cell that both triggers and coordinates key events in the cell cycle.
- cell division** The reproduction of cells.
- cell fractionation** The disruption of a cell and separation of its parts by centrifugation at successively higher speeds.
- cell-mediated immune response** The branch of adaptive immunity that involves the activation of cytotoxic T cells, which defend against infected cells.
- cell plate** A membrane-bounded, flattened sac located at the midline of a dividing plant cell, inside which the new cell wall forms during cytokinesis.
- cellular respiration** The catabolic pathways of aerobic and anaerobic respiration, which break down organic molecules and use an electron transport chain for the production of ATP.
- cellulose** (sel'-yū-lōs) A structural polysaccharide of plant cell walls, consisting of glucose monomers joined by β glycosidic linkages.
- cell wall** A protective layer external to the plasma membrane in the cells of plants, prokaryotes, fungi, and some protists. Polysaccharides such as cellulose (in plants and some protists), chitin (in fungi), and peptidoglycan (in bacteria) are important structural components of cell walls.
- central nervous system (CNS)** The portion of the nervous system where signal integration occurs; in vertebrate animals, the brain and spinal cord.
- central vacuole** In a mature plant cell, a large membranous sac with diverse roles in growth, storage, and sequestration of toxic substances.
- centriole** (sen'-trē-ōl) A structure in the centrosome of an animal cell composed of a cylinder of microtubule triplets arranged in a "9 + 0" pattern. A centrosome has a pair of centrioles.
- centromere** (sen'-trō-mēr) In a duplicated chromosome, the region on each sister chromatid where it is most closely attached to its sister chromatid by proteins that bind to the centromeric DNA. Other proteins condense the chromatin in that region, so it appears as a narrow "waist" on the duplicated chromosome. (An unduplicated chromosome has a single centromere, identified by the proteins bound there.)
- centrosome** (sen'-trō-sōm) A structure present in the cytoplasm of animal cells that functions as a microtubule-organizing center and is important during cell division. A centrosome has two centrioles.
- cerczoan** An amoeboid or flagellated protist that feeds with threadlike pseudopodia.
- cerebellum** (sār'-ruh-bel'-um) Part of the vertebrate hindbrain located dorsally; functions in unconscious coordination of movement and balance.
- cerebral cortex** (suh-rē'-brul) The surface of the cerebrum; the largest and most complex part of the mammalian brain, containing nerve cell bodies of the cerebrum; the part of the vertebrate brain most changed through evolution.
- cerebrum** (suh-rē'-brum) The dorsal portion of the vertebrate forebrain, composed of right and left hemispheres; the integrating center for memory, learning, emotions, and other highly complex functions of the central nervous system.
- cervix** (ser'-viks) The neck of the uterus, which opens into the vagina.
- chaparral** A scrubland biome of dense, spiny evergreen shrubs found at midlatitudes along coasts where cold ocean currents circulate offshore; characterized by mild, rainy winters and long, hot, dry summers.
- chaperonin** (shap'-er-ō'-nin) A protein complex that assists in the proper folding of other proteins.
- character** An observable heritable feature that may vary among individuals.
- character displacement** The tendency for characteristics to be more divergent in sympatric populations of two species than in allopatric populations of the same two species.
- checkpoint** A control point in the cell cycle where stop and go-ahead signals can regulate the cycle.
- chelicera** (kē-lih'-suh-ruh) (plural, **chelicerae**) One of a pair of clawlike feeding appendages characteristic of chelicerates.
- chelicerate** (kē-lih-suh'-rāte) An arthropod that has chelicerae and a body divided into a cephalothorax and an abdomen. Living chelicerates include sea spiders, horseshoe crabs, scorpions, ticks, and spiders.
- chemical bond** An attraction between two atoms, resulting from a sharing of outer-shell electrons or the presence of opposite charges on the atoms. The bonded atoms gain complete outer electron shells.
- chemical energy** Energy available in molecules for release in a chemical reaction; a form of potential energy.
- chemical equilibrium** In a chemical reaction, the state in which the rate of the forward reaction equals the rate of the reverse reaction, so that the relative concentrations of the reactants and products do not change with time.
- chemical reaction** The making and breaking of chemical bonds, leading to changes in the composition of matter.
- chemiosmosis** (kem'-ē-oz-mō'-sis) An energy-coupling mechanism that uses energy stored in the form of a hydrogen ion gradient across a membrane to drive cellular work, such as the synthesis of ATP. Under aerobic conditions, most ATP synthesis in cells occurs by chemiosmosis.
- chemoautotroph** (kē'-mō-ot'-ō-trōf) An organism that obtains energy by oxidizing inorganic substances and needs only carbon dioxide as a carbon source.
- chemoheterotroph** (kē'-mō-het'-er-ō-trōf) An organism that requires organic molecules for both energy and carbon.
- chemoreceptor** A sensory receptor that responds to a chemical stimulus, such as a solute or an odorant.
- chiasma** (plural, **chiasmata**) (kī-az'-muh, kī-az'-muh-tuh) The X-shaped, microscopically visible region where crossing over has occurred earlier in prophase I between homologous nonsister chromatids. Chiasmata become visible after synapsis ends, with the two homologs remaining associated due to sister chromatid cohesion.
- chitin** (kī'-tin) A structural polysaccharide, consisting of amino sugar monomers, found in many fungal cell walls and in the exoskeletons of all arthropods.
- chlorophyll** (klōr'-ō-fil) A green pigment located in membranes within the chloroplasts of plants and algae and in the membranes of certain prokaryotes. Chlorophyll *a* participates directly in the light reactions, which convert solar energy to chemical energy.
- chlorophyll *a*** (klōr'-ō-fil) A photosynthetic pigment that participates directly in the light reactions, which convert solar energy to chemical energy.
- chlorophyll *b*** (klōr'-ō-fil) An accessory photosynthetic pigment that transfers energy to chlorophyll *a*.
- chloroplast** (klōr'-ō-plast) An organelle found in plants and photosynthetic protists that absorbs sunlight and uses it to drive the synthesis of organic compounds from carbon dioxide and water.
- choanocyte** (kō-an'-uh-sit) A flagellated feeding cell found in sponges. Also called a collar cell, it has a collar-like ring that traps food particles around the base of its flagellum.
- cholesterol** (kō-les'-tuh-rol) A steroid that forms an essential component of animal cell membranes and acts as a precursor molecule for the synthesis of other biologically important steroids, such as many hormones.
- chondrichthyan** (kon-drik'-thē-an) A member of the clade Chondrichthyes, vertebrates with skeletons made mostly of cartilage, such as sharks and rays.
- chordate** A member of the phylum Chordata, animals that at some point during their development have a notochord; a dorsal, hollow nerve cord; pharyngeal slits or clefts; and a muscular, post-anal tail.
- chorionic villus sampling (CVS)** (kōr'-ē-on'-ik vil'-us) A technique associated with prenatal diagnosis in which a small sample of the fetal portion of the placenta is removed for analysis to detect certain genetic and congenital defects in the fetus.
- chromatin** (krō'-muh-tin) The complex of DNA and proteins that makes up eukaryotic chromosomes. When the cell is not dividing, chromatin exists in its dispersed form, as a mass of